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STUDY AND EVALUATION SCHEME
(With effective from academic session 2023-2024)
BTech. in Cloud Computing
YEAR III, SEMESTER V

Sl. No.	Category	Course Code	Course Title/ Subjects	Hours per week			Evaluation Scheme		Total	Credits
				L	T	P	CA	EE		
THEORY										
1	Professional Core	BCSI501	Theory of Computation	3	1	0	30	70	100	4
2	Professional Core	BCSICT501	Cloud Computing	3	0	0	25	50	75	3
3	Professional Core	BCSICT502	Network Security	3	0	0	25	50	75	3
4	Professional Core	BCSICT503	Principles of Virtualization	1	0	0	10	15	25	1
5	HSM	BCSICT504	Humanities II	3	1	0	30	70	100	4
6	Professional Elective		Elective-I	3	0	0	25	50	75	3
7	Engineering Science Course	IIOT5	Deep Learning	4	0	0	30	70	100	4
PRACTICALS AND PROJECTS										
7	Professional Core	BCSICT508	Network Security Lab	0	0	2	10	15	25	1
8	Professional Core	BCSICT509	Principle of Virtualization Lab	0	0	2	20	30	50	2
9	Summer Training	BCSI502	Summer Project Seminar-II	0	0	2	20	30	50	2
			TOTAL	20	2	6	225	450	625	27

L-Lecture, **T**- Tutorial , **P**- Practical ,**CA**- Continuous Assessment, **EE**- End Semester Examination

Cloud Computing

L*-Lecture* , T*-Tutorial, P*-Practical

COURSE CODE	Elective - I
BCSICT505	Security Architecture
BCSICT506	Database Security

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STUDY AND EVALUATION SCHEME
(With effective from academic session 2023-2024)
B.Tech. in Cloud Computing
YEAR III, SEMESTER VI

Sl. No.	Category	Course Code	Course Title/ Subjects	Hours per week			Evaluation Scheme		Total	Credits
				L	T	P	CA	EE		
THEORY										
1	Professional Core	BCSI601	Artificial Intelligence	3	0	0	25	50	75	3
2	Professional Core	BCSICT601	Linux Administration	1	0	0	10	15	25	1
3	Professional Core	BCSICT602	Ethical Hacking	3	1	0	30	70	100	4
4	Professional Core		Elective -II	3	1	0	30	70	100	4
5	Professional Core	BCSICT608	Incident Response Management	2	0	0	15	35	50	2
6	Open Elective		Open Elective-I	3	0	0	25	50	75	3
7	Engineering Science Course	IIOT6	Advanced Artificial Intelligence	4	0	0	30	70	100	4
PRACTICALS AND PROJECTS										
7	Professional Core	BCSICT612	Exploring Software as a Service (SaaS) Lab	0	0	2	20	30	50	2
8	Professional Core	SICT615	Linux administration Lab	0	0	2	20	30	50	2
9	Professional Core	SICT616	Ethical Hacking Lab	0	0	2	10	15	25	1
10	Project	BCSICT617	Project-I			4				
			TOTAL	19	2	12	215	435	650	26

L-Lecture, **T**- Tutorial , **P**- Practical ,**CA**- Continuous Assessment, **EE**- End Semester Examination

Course code	Open Elective - I
BCSICT609	UI/UX Fundamentals
SICT610	Mobile Application Development
BCSICT611	Business Intelligence

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BCSICT501 Cloud Computing

L T P C
3 0 0 3

MODULE-I

Introduction to Cloud Computing–Definition of Cloud, Evolution of Cloud computing, Underlying Principles of Parallel and Distributed Computing ,Cloud Characteristics , Elasticity in Cloud, On-demand Provisioning. Service Oriented Architecture, REST and Systems of Systems Web Services, Publish-Subscribe Model, Basics of Virtualization, Types of Virtualization Implementation Levels of Virtualization, Virtualization Structures Tools And Mechanisms, Virtualization of CPU, Memory I/O Devices, Virtualization Support and Disaster Recovery

MODULE-II

Layered Cloud Architecture Design – NIST Cloud Computing Reference Architecture – Public, Private and Hybrid Clouds laaS, PaaS, SaaS, Architectural Design Challenges, Cloud Storage Storages-a-Service, Advantages of Cloud Storage ,Cloud Storage ProvidersS3.Need for Virtualization Pros and cons of Virtualization Types of Virtualization –System VM, Process VM, Virtual Machine monitor – Virtual machine properties - Interpretation and binary translation, HLL VM - supervisors Xen, KVM, VMware, Virtual Box, Hyper-V, Major Players in Cloud Computing issues in Clouds - Eucalyptus - Nimbus - Open Nebula, CloudSim

MODULE-III

Security Standards. Security, Standards and Applications: Security in Clouds: Cloud security challenges – Software as a Service Security, Common Standards: The Open Cloud Consortium – The Distributed management Task Force – Standards for application Developers – Standards for Messaging – Standards for Security, End user access to cloud computing, Mobile Internet devices and the cloud. Service providers- Google, Amazon, Microsoft Azure, IBM, Sales force

Text Books:

1. Kai Hwang, Geoffrey C. Fox, Jack G. Dongarra, "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers, 2012.
2. Rittinghouse, John W., and James F. Ransome, ―Cloud Computing: Implementation, Management and Security, CRC Press, 2017.
3. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, ―Mastering Cloud Computing, Tata Mcgraw Hill, 2013.
4. Toby Velte, Anthony Velte, Robert Elsenpeter, "Cloud Computing – A Practical Approach, Tata Mcgraw Hill, 2009.
5. George Reese, "Cloud Application Architectures: Building Applications and Infrastructure in the Cloud: Transactional Systems for EC2 and Beyond (Theory in Practice), O'Reilly, 2009.

BCSICT503 - Principles of Virtualization

L T P C

1 0 0 1

Course Objective:

This course gives students an insight into the basics of cloud computing along with virtualization, Cloud computing is one of the fastest growing domain from a while now. It will provide the students Basic understanding about cloud and virtualization along with it how one can migrate over it

MODULE-I

Introduction to Virtualization: Virtualization and cloud computing - Need of virtualization – cost, administration, fast deployment, reduce infrastructure cost – limitations Types of hardware virtualization: Full virtualization - partial virtualization - Para virtualization
Desktop virtualization: Software virtualization – Memory virtualization - Storage virtualization – Data virtualization – Network virtualization)

MODULE-II

Hypervisors and Virtual machines Server Virtualization: Understanding Server Virtualization, types of server virtualization, Virtual machine basics, types of virtual machines, hypervisor concepts and types Virtualization Solutions Understanding Microsoft's Virtualization solutions: Microsoft's Infrastructure Optimization Model, Virtualization and the Infrastructure Optimization Model, Benefits of Virtualization, Achieving the Benefits of Datacenter Virtualization, Achieving the Benefits of Client Virtualization, Achieving the Benefits of Cloud Virtualization

MODULE-III

Migrating into a Cloud Introduction, Challenges while migrating to Cloud, Broad approaches to migrating into the cloud why migrate -deciding on cloud migration, the Seven-step model of migration into a cloud, Migration Risks and Mitigation, Enterprise cloud computing paradigm, relevant Deployment Models for Enterprise Cloud Computing, Adoption and Consumption Strategies, issues for enterprise applications on the cloud

Text & References:

Text:

1. David Marshall, Wade A. Reynolds, Advanced Server Virtualization: VMware an Microsoft Platform in the Virtual Data Center, Acerbic
2. Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online - Michael Miller - Que 2008
3. Cloud Computing (Principles and Paradigms), Edited by Rajkumar Buyya, James Broberg Andrzej Goscinski, John Wiley & Sons, Inc. 2011
4. Cloud computing a practical approach - Anthony T.Velte, Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010

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BCSICT504 – Humanities II Engineering & Managerial Economics

MODULE-I

Introduction: Meaning, Nature and Scope of Economics, Meaning of Science, Engineering and Technology. Managerial Economics and its scope in engineering perspective.

MODULE-II

Basic Concepts Demand Analysis, Law of Demand, Determinates of Demand, Elasticity of Demand-Price, Income and cross Elasticity. Uses of concept of elasticity of demand in managerial decision

MODULE-III

Demand forecasting Meaning, significance and methods of demand forecasting, production function, Laws of returns to scale & Law of Diminishing returns scale. An overview of Short and Long run cost curves – fixed cost, variable cost, average cost, marginal cost, Opportunity cost

MODULE-IV

Market Structure Perfect Competition, Imperfect competition – Monopolistic, Oligopoly, duopoly sorbent features of price determination and various market conditions.

MODULE-V

National Income, Inflation and Business Cycles Concept of N.I. and Measurement. Meaning of Inflation, Type causes & prevention methods, Phases of business cycle

Reference Books:-

1. Koutsoyiannis A: Modern Microeconomics, ELBS.
2. Managerial Economics for Engineering: Prof. D.N. Kakkar
3. Managerial Economics: D.N. Dwivedi
4: Managerial Economics: Maheshwari.

Elective-I

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BCSICT505 Security Architecture

MODULE-I

Introduction to Cloud Computing and Security: Understanding Cloud Computing - The IT Foundation for Cloud- overview of Security Architecture, Cloud Computing Architecture: loud Reference Architecture-Control over Security in the Cloud Model- Cloud Deployment & Services Models- Key Examples

MODULE-II

Cloud Computing: Security Concerns- Risk Tolerance- Legal and Regulatory Issues, Security Requirements for the Architecture-Security Patterns and Architectural Elements- Cloud Security Architecture-Key Strategies for Secure Operation

MODULE-III

Overview of Data Security in Cloud Computing-Common Risks with Cloud Data Security- Data Encryption: Applications and Limits- Errors with Data Encryption- Cloud Data Security: Sensitive Data Categorization, Cloud Data Storage-Roach Motel Syndrome, Overall Strategy: Effectively Managing Risk, Overview of Security Controls, Overview of Security Controls, The Limits of Security Controls,
Best Practices, Security Monitoring

MODULE-IV

Private Clouds: Motivation and Overview-Security Implications: Shared versus Dedicated Resources, Security Criteria for Ensuring a Private Cloud - Network Considerations- Data Center Considerations- Operational Security Considerations- Regulation, Selecting a CSP: Overview of Assurance, Overview of Risks, and Security Criteria- Revisiting Defense-in-depth- Additional Security relevant Criteria.

MODULE-V

Evaluating Cloud Security, Checklists for Evaluating Cloud Security- Foundational Security- Business Considerations- Defense-in-depth- Operational Security, Operating a Cloud: From Architecture to Efficient and Secure Operations, Bootstrapping Secure Operations, Security Operations Activities- Business Continuity, Backup, and Recovery- Managing Changes in Operational Environments - Information Security Management - Vulnerability and Penetration Testing, Security Monitoring and Response

TEXT BOOKS

Vic (J.R.) Winkler, “Securing the Cloud: Cloud Computer Security Techniques and Tactics”, Elsevier,2011.

REFERENCE BOOKS

1. Sushil Jajodia, Krishna Kant, “Secure Cloud Computing”, Elsevier,2014. 2. Curtis Franklin, Jr. ,Brian J. S. Chee, “Securing the Cloud: Security Strategies for the Ubiquitous Data Center”, CRC Press, 2019.

EBOOk

1. https://solutionsreview.com/cloud-platforms/free-cloud-computing-ebooks/

MOOC

1 https://www.coursera.org/learn/cloud-computing-security

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BCSICT602 PC Ethical Hacking

MODULE-I

Cyber Ethical Hacking ,What is Cyber Attack,Types of Cyber Attack,Phase of hacking,Information Gathering,Scanning,Google Hacking Database,type of Virus,type of Worms ,Virus ,Trojans and Backdoors,Sniffers and keyloggers,Social Engineering,Email, DNS, IP spoofing ,System Hacking and Security,HoneyPots

MODULE-II

Footprinting and Reconnaissance ,Scanning Networks,Enumeration,System Hacking,Malware Threats,Sniffing,Social Engineering, Denial of Service, Session Hijacking Hacking Web Servers, Hacking Web Applications,SQL Injection ,Hacking Wireless Networks, Hacking Mobile Platforms, Evading IDS, Firewalls and Honey pot

MODULE-III

Ethical Hacking Cloud Computing, Ethical Hacking Cryptography, Denial of Service, Hacking Mobile Platforms, Hacking Web Servers, Session Hijacking, Scanning Networks, Social Engineering, Malware Threats,Footprinting and Reconnaissance,SQL Injection, Evading IDS, Firewalls and Honey pots Enumeration,IoT and OT Hacking

Reference Books:-

1. Hands-on Ethical Hacking and Network Defense.
2. The Basics of Hacking and Penetration Testing - Patrick Engebretson.
3. The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws.
4. Black Hat Python: Python Programming for Hackers and Pentesters.
5. Hacking: The Art of Exploitation by Jon Erickson.

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BCSICT605 Cloud Scripting using PaaS

COURSE DESCRIPTION

This course provides a hands-on comprehensive study of Cloud concepts and capabilities across the various Cloud service models including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Business Process as a Service (BPaaS). IaaS topics start with a detailed study the evolution of infrastructure migration approaches from VMWare/Xen/ KVM virtualization, to adaptive virtualization, and on-demand resources provisioning. PaaS topics cover a broad range of Cloud vendor platforms including Google App Engine, Microsoft Azure, OpenStack and others as well as a detailed study of related platform services such as storage services that leverage Google Storage, Amazon S3, Amazon Dynamo, or other services meant to provide Cloud resources management and monitoring capabilities. The SaaS and PaaS topics covered in the course will familiarize students with the use of vendor-maintained applications and processes available on the Cloud on a metered on-demand basis in multi-tenant environments. The course also covers the Cloud security model and associated challenges and delves into the implementation and support of High Performance Computing and Big Data support capabilities on the Cloud. Through hands-on assignments and projects, students will learn how to configure and program IaaS services.

COURSE OBJECTIVES

learn cloud computing delivery model IaaS
learn cloud computing delivery model PaaS
learn cloud computing delivery model SaaS.

COURSE OUTCOMES

On completion of this course, the students will be able to
understand Cloud delivery models in details
understand briefly Cloud Computing Reference Architecture.

MODULE-I

Introduction of delivery models in Cloud Computing: Introduction to cloud delivery models, List various cloud delivery models, Advantages of delivery models in cloud, trade-off in cost to install versus flexibility, Cloud service model architecture.

MODULE-II

Infrastructure as a Service (IaaS): Introduction to Infrastructure as a Service delivery model, characteristics of IaaS, Architecture, examples of IaaS, Applicability of IaaS in the industry.

MODULE-III

Platform as a Service (PaaS):Introduction to Platform as a Service delivery model, characteristics of PaaS, patterns, architecture and examples of PaaS, Applicability of PaaS in the industry

MODULE-IV

Software as a Service (SaaS):Introduction to Software as a Service delivery model, characteristics of SaaS, Architecture, examples of SaaS, Applicability of SaaS in the industry..

MODULE-V

Cloud computing Reference Architecture (CCRA): Introduction to Cloud computing reference architecture (CCRA), benefits of CCRA, Architecture overview, versions and application of CCRA for developing clouds, Type causes & prevention methods, Phases of business cycle.

Cengage Learning India Pvt Ltd, 2009
9. Crafting the InfoSec Playbook: Security Monitoring and Incident Response Master Plan 1st Kindle Edition by Jeff Bollinger, Brandon Enright, Matthew Valites, 2015
10. Incident Management and Response Guide: Tools, Techniques, Planning, and Templates Kindle Edition by Tom Olzak, Erudio Security, 2017
11. Cyber Security by Nina Godbole, SunitBelapure, Wiley, 2011
12. Incident Response & Computer Forensics by Jason T. Luttgens, Matthew Pepe, Kevin Mandia, McGraw-Hill Education; 3rd edition, 2014
13. Principles of Incident Response and Disaster Recovery by Michael Whitman, Herbert MattordDelmar Cengage Learning; 2nd Revised edition, 2013
14. Computer Incident Response and Forensics Team Management: Conducting a Successful Incident Response by Leighton Johnson, Syngress, 2013
15. Cyber Incident Response: Bridging the Gap Between Cybersecurity and Emergency Management by Response, and Communications and the Subcommittee on Cybersecurity, Infrastructure Protection, and Security Technologies of the Committee on Homeland Security House of Representatives Subcommittee on Emergency Preparedness, CreateSpace Independent Publishing Platform (8 May 2014)

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Open Elective-I

BCSICT609 UI/UX Fundamentals

Course Description: The increasing possibilities with interactive technology as opened to virtual classrooms for teaching and educating the students. Research has proven that interactive teaching using such visual technologies is much more effective than the traditional methods which help students understand and gain knowledge better. Virtual reality is used in many training scenarios as it consists of a wide range of benefits for academia and industrial needs. Course Objectives: Students will get to know about various techniques of Graphic Design and UI/UX and will develop skills to become a professional designer. They will be taught to enhance their knowledge and master tools producing good industry standard designs. Students will be able to work on advertisements, website, and app designs.

Course Outcomes (COs): At the end of this course students will be able to:
CO 1. Create Graphic Design artworks of your own.
CO 2. Explain the functionality of different design related software
CO 3. Use learned skills to solve problems of various layouts
CO 4. Test own's skill and knowledge for a better workflow
CO 5. Select best output and what works for a particular given project
CO 6. Develop ideas and various app designs and website pages..

MODULE-I

Unit 1: Computer Fundamentals & Digital Illustration, Introduction to Graphic Design and Its Uses, Raster & Vector Graphics, Drawing Vector Shapes and Illustrations, Art & Sketching, Drawing Techniques, Conceptual Thinking in Creativity, Developing a Personal Illustration Style, Color Modes, Schemes, Design, Image Retouching and Color Balancing, Using Filters Corporate Identity Design Designing Brochures & Catalogues Layouts for Newspapers, Designing Magazines, Visual Design Principles

MODULE-II

UI design fundamentals, Evolution of user interfaces, Interaction with physical components, Flat design, Role of UI in UX,Laws of digital interface design, Understand user experience,VIMM model, Know your user, user research, Difference between design and art, emotional design, designing for mental models, Importance of presentation, content, interactions, screen elements, accessibility, and global standards

MODULE-III

Typography, Types of typefaces, Typography Terminology, Guidelines for proper type selection, Typography design, Analyzing Aesthetics as per laws of Design principles, Alignment, Spacing, Lighting & Shadows, Grids,Consistency

MODULE-IV

Design Language & Rapid Prototyping • In-depth study of Design languages for different Google material platforms • Style guides and its importance • Apple Human centered Design guidelines measurements of UI components • Design for platforms: Mobile, Web, Tablet, Responsive, Smart Watch • Mood boards • UX design principles and laws • Introduction to Adobe XD • Creating low and high-fidelity prototypes

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BCSICT610 Mobile ApplicationDevelopment

MODULE-I

Introduction to Android: The Android Platform, Android SDK, Eclipse Installation, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file..

MODULE-II

Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.

MODULE-III

Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts, Drawing and Working with Animation.

MODULE-IV

Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources. UNIT - V Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using Android Web APIs, Using Android Telephony APIs, Deploying Android Application to the World..

TEXT BOOKS: 1.

T1. Lauren Darcey and Shane Conder, “Android Wireless Application Development”, Pearson Education, 2nd ed. (2011)

REFERENCE BOOKS:

R1. Reto Meier, “Professional Android 2 Application Development”, Wiley India Pvt Ltd
R2. Mark L Murphy, “Beginning Android”, Wiley India Pvt Ltd
R3. Android Application Development All in one for Dummies by Barry Burd, Edition: I

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BCSICT616 PC Ethical Hacking Lab

Course Objectives:

Introduces the concepts of Ethical Hacking

Gives the students the opportunity to learn about different tools and techniques in Ethical hacking and security Practically apply Ethical hacking tools to perform various activities.

Course Outcomes:

After completion of course, students would be able to:

Understand the core concepts related to vulnerabilities and their causes

Understand ethics behind hacking and vulnerability disclosure

Appreciate the impact of hacking

Perform the lab Experiments based on the following facts.(Develop your own set of lab experiments)

1 Google Hacking

2 Scanning

- Locating Open Ports
- Network mapping
- OS Fingerprinting

3 Gaining and Maintaining Access
4 Passwords hacking
5 Password Cracking Methods
6 Password Cracking Software
7 Man-in-the-Middle
8 Backdoors
9 Denial of Service
10 Covering tracks
11 Intrusion Detection Systems
12 Intrusion Prevention Systems
13 Anti-viruses
14 Malware
15 Viruses